

Voicelsolate Pro

v24.0.0 · Latest Product & Engineering Snapshot

100% on-device voice isolation · Analyzer ↔ WhisperHunter · Web · Desktop · Android

This document summarizes the current production product (Engineer Mode + Stem-Split), the deferred-decode freeze-resistant pipeline, Analyzer↔WhisperHunter collaboration, and platform packaging for browser, Electron desktop, and Capacitor Android.

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1. Product overview

Voicelso Pro extracts studio-quality voice from noisy recordings without sending audio to a server. Processing is local: classical DSP + ONNX Runtime Web (WebGPU preferred, WASM fallback). Live microphone capture is intentionally disabled (upload-only).

Surface	Role
Landing /	Fast ML stem separation (vocals / accompaniment / noise)
Engineer /app/	Full DSP suite, analysis workspace, WhisperHunter, Live-Mix
Desktop	Electron shell + native save dialogs
Android	Capacitor WebView, offline models, COOP/COEP hardening

2. Engineer Mode workflow (v24)

Upload → **Analyze** → **Isolate** is the disciplined path. Decode is deferred so large files no longer freeze the browser at drop time.

- **Upload** — accept File only (name/size/type). Optional video picture via blob URL. No PCM decode.
- **Analyze Full Audio** — *ensureDecoded()* then full-file classical analysis (speech, whisper, music, noise, hum, impulses).
- **Joint map** — AnalyzerWhisperBridge fuses analysis with WhisperHunter environment profiling: protect voice/whisper; suppress music, horns, barks, crowd, hum, reverb.
- **Analyze + WhisperHunter** — one-shot apply joint plan + single-pass isolation.
- **Process / Live-Mix** — ML stem path when available, single STFT→ops→iSTFT spectral chain, then real-time gate/de-ess/EQ preview.
- **Export** — WAV (or remuxed video with processed audio).

3. Freeze resistance & speed

- Deferred decode on Analyze / Process / Play (not at upload).
- No auto-pipeline on file drop — user starts work when ready.
- Cooperative yields: *scheduler.yield* when available, else rAF + macrotask; budgeted STFT yields.
- Stereo mid-channel process path halves spectral cost.
- Live-Mix AudioWorklets load lazily — never block upload or decode.
- Single-pass spectral contract — no multi-pass STFT freeze loops.

4. Analyzer ↔ WhisperHunter collaboration

The analysis workspace and WhisperHunter AI share one isolation map so unwanted sounds are suppressed while voices and whispers are protected and amplified (no cloud ASR, no invented words).

Class	Response
Music bed	musicKill, voicelso, bassCrush
Impulses (horns, barks)	crowdNull + NR; protect speech formants
Broadband / traffic / crowd	NR, HP, isolation

Hum / reverb	humRemoval, dereverb, roomCorrection
Whisper / faint speech	shallower gate, whisperLift, presence boost

5. Platforms

Web: Vercel + COOP/COEP for SharedArrayBuffer · **Desktop:** Electron · **Android:** Capacitor, MainActivity
COOP/COEP injection, lean offline model pack (BSRNN + RNNoise + VAD).

6. Non-negotiable architecture constraints

- Exactly one forward STFT and one inverse STFT per offline spectral branch.
- 100% local audio processing — no cloud inference of user media.
- Upload-only — getUserMedia forbidden.
- Honest audition quality badges (high / medium / low).
- Models same-origin under /app/models/ with integrity checks when shipped.

Canonical code contract: CLAUDE.md · Architecture: docs/architecture/ · Guides: docs/guides/ · Repository:
<https://github.com/Joker5514/VoiceIsolate-Pro>

Document generated for the latest v24 production line — Engineer Mode collaboration + deferred decode.